

CSC5025 - Intelligent Systems

Assignment 2: Spatial Temporal Graph Neural Networks

Maximum marks - 30

Objective

This assignment explores Spatial Temporal Graph Neural Networks (STGNNs) for predictive modelling using a real world experimental platform. You will learn to develop and analyse ML experiments using STGNNs, specifically GraphWaveNet (GWN) and the Adaptive Graph Convolutional Recurrent Network (AGCRN) for predictive modelling of traffic patterns on the full METR-LA data set.

Learning outcomes

- Building on the paper discussion task on STGNNs, this assignment will allow you to gain both a theoretical and practical understanding of key concepts, established practice and tools for implementing and analysing STGNNs.
- Learn how to configure and use a current real-world STGNN library and experimental platform for sequence to sequence predictions
 - Logging, model checkpoints, experimental pipelines
- Understand how the adjacency matrix encodes spatial temporal dependencies between traffic nodes and how this can be learned from the data
- Analyse and compare GWN and AGCRN in terms of predictive performance, convergence, training time and spatial modelling.

Task

Using the Torch Spatiotemporal library implement and analyse the simple TimeThenSpaceModel, GWN and the AGCRN for traffic forecasting on the MetrLA data set. You should use the full MetrLA data set provided in the TSL library.

Use the following settings for all experiments:

- Holdout validation using the partitioning ratio: 0.7 : 0.1 : 0.2
- Standard scaler
- Window size = 12 (60 minutes)
- Prediction horizon = 12 (60 minutes)
- Report performance at prediction horizons of 15, 30 and 60 minutes
- Use the same data split, evaluation metrics and hardware environment when comparing models

Question 1 (6 marks)

Activities

- Using the TSL workflow introduced in the tutorial, configure and run the baseline TimeThenSpaceModel on the full METRLA dataset.

Report

1. Provide a heatmap to visualise the adjacency matrix.
2. Average performance for the baseline TimeThenSpaceModel (TTS) across all stations: Provide a table and a graph to show the average MSE, MAPE and MAE across all 207 sensors for different prediction horizons (15, 30 and 60 minutes).
3. Per station analysis: Using appropriate graphs, analyse the actual versus predicted readings for the test set on the first 3 sensors.

Question 2 (14 marks)

Activities

- Repeat the experiment from Question 1 using GraphWaveNet. Use the default GraphWaveNet hyperparameters provided in the TSL library.
- Run the following two GraphWaveNet configurations:
 - i. GraphWaveNet with a predefined graph only
 - ii. GraphWaveNet with a predefined graph and learned adaptive adjacency

Report

1. Average overall performance across all stations
 - Compare the two GWN configurations with the simple TTS in terms of training time, convergence over training steps and average performance on all three metrics (MSE, MAPE and MAE). You may use TensorBoard for your investigations.
 - Discuss whether the learned adaptive adjacency improves performance
 - Compare your GWN results with those reported in the original GraphWaveNet paper [1] using directly comparable metrics. Identify any differences that may limit direct comparison.
2. Per station analysis
 - Compare test performance on first three nodes across all three metrics MSE, MAPE and MAE for different prediction horizons, 15 minute, 30 minute and 60 minute
 - Describe the effect of the learned adaptive adjacency and whether it is consistent across the first 3 sensors.
3. Learned adjacency analysis:
 - Extract and analyse the adjacency matrix for GraphWaveNet for the first 50 nodes.
 - Determine the top 15 most influential nodes and which nodes they most impact on.
 - Compare the learned adaptive adjacency matrix of GWN with the one used in the TTS experiment

Question 3 (10 marks)

Activities

- Repeat the experiments from Question 1 using AGCRN [2]. You should do some initial experiments to determine the optimal number of epochs to run for AGCRN. Use the default AGCRN hyperparameters provided in the TSL library.

Report

Compare and analyse the Adaptive Graph Convolutional Recurrent Network (AGCRN) with GWN in terms of:

1. Average overall performance in terms of error metrics (MSE, MAPE and MAE), training time and convergence.
2. Per station analysis using the first three nodes.
3. Provide a final summary analysis of your findings, specifically of how AGCRN compares to GWN for the traffic prediction task.
4. Compare and reflect on your findings with the results and findings in [4], which explored both AGCRN and GWN for weather prediction.

Useful links:

- Torch Spatiotemporal - <https://torch-spatiotemporal.readthedocs.io/en/latest/>

References

1. Wu, Z., Pan, S., Long, G., Jiang, J. and Zhang, C., 2019. Graph wavenet for deep spatial-temporal graph modeling. arXiv preprint arXiv:1906.00121. Bai, L., Yao, L., Li, C., Wang, X. and Wang, C., 2020. Adaptive graph convolutional recurrent network for traffic forecasting. *Advances in neural information processing systems*, 33, pp.17804-17815.
2. Bai, L., Yao, L., Li, C., Wang, X. and Wang, C., 2020. Adaptive graph convolutional recurrent network for traffic forecasting. *Advances in neural information processing systems*, 33, pp.17804-17815.
3. Cini, A., Marisca, I., Zambon, D. and Alippi, C., 2023. Graph deep learning for time series forecasting. arXiv preprint arXiv:2310.15978.
4. Gaibie, Adeeb, Hamza Amir, Irene Nandutu, and Deshendran Moodley. "Predicting and Discovering Weather Patterns in South Africa Using Spatial-Temporal Graph Neural Networks." In *Southern African Conference for Artificial Intelligence Research*, pp. 144-160. Cham: Springer Nature Switzerland, 2024.

Submission

Each student must submit a single zipped archive file using the naming convention [student id][surname][initials].zip.

The file should contain:

- Pytorch code - in a folder called "code"
- A single PDF report file in the root folder
 - [student id][surname][initials].pdf